

Project Title: Exploratory Data Analysis (EDA) on Online Casino Gaming Revenue

Domain: Sports

Difficulty Level: Intermediate

Use the **Game Data** from Kaggle: [Dataset Link](#)

Background

As a data science trainee, you have been hired as a data analyst for a consulting firm specializing in the gaming industry. Your firm has been contracted by a consortium of online casinos to analyze their performance metrics and identify trends, patterns, and potential areas of improvement.

Your task is to perform **Exploratory Data Analysis (EDA)** on a dataset containing detailed monthly reports from different online casino gaming licenses. Your findings will help casino operators make informed business and regulatory decisions.

Objective

The goal of this assignment is to:

1. **Analyze** trends in online casino gaming revenue across different licenses and periods.
2. **Identify** key performance metrics, such as net winnings, promotional deductions, and gross revenue.
3. **Discover** correlations between variables, such as promotional activity and total revenue.
4. **Handle** missing values and outliers to ensure a robust analysis.
5. **Provide** data-driven insights and recommendations to casino operators.

Dataset Details

Format: CSV

Columns:

- **Licensee:** Name of the casino license holder
- **Fiscal Year:** The fiscal year of the report
- **Month Ending:** The month for which the report is prepared
- **Wagers:** Total amount wagered by patrons
- **Patron Winnings:** Total winnings paid to patrons
- **Cancelled Wagers:** Wagers that were cancelled
- **Online Casino Gaming Win/(Loss):** Net win or loss from online gaming
- **Promotional Coupons or Credits Wagered:** Amount wagered using promotional credits

- **Promotional Deduction:** Total amount deducted due to promotions
 - **Total Gross Gaming Revenue:** Total revenue from gaming activities
 - **Payment:** Total payment to the license holder
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Scenario & Problem Statements

Scenario

An online casino consortium has noticed fluctuations in its revenue and wants to understand the underlying causes. They want insights into which platforms perform best, how promotions impact revenue, and how different casinos compare in performance.

Problem Statements

1. **Revenue Trends Analysis:**
 - How has the total gaming revenue changed over time?
 - Are there seasonal trends in casino revenue?
 2. **Impact of Promotions:**
 - Do promotional credits and deductions lead to higher overall revenue?
 - Is there a correlation between promotional spending and net winnings?
 3. **Casino Performance Comparison:**
 - Which casino licenses generate the highest revenue?
 - Are there specific casinos consistently underperforming?
 4. **Wagering and Payout Patterns:**
 - How do cancelled wagers impact revenue?
 - What percentage of wagers is typically returned as winnings?
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Assignment Instructions

1. **Data Preprocessing**
 - Load and inspect the dataset for missing values and outliers.
 - Handle missing values using appropriate techniques (e.g., imputation, removal).
 - Check for duplicate entries and correct them if necessary.
2. **Exploratory Data Analysis (EDA)**
 - **Descriptive Statistics:** Calculate mean, median, standard deviation, and other summary statistics.
 - **Visualizations:**

- Plot histograms for revenue distribution.
- Use line charts to observe revenue trends over time.
- Create scatter plots to analyze relationships between variables.
- Generate bar charts to compare casino performance.

3. Correlation & Insights

- Compute correlation matrices to identify strong relationships between variables.
- Investigate whether higher promotional spending leads to increased revenue.
- Compare different casino licenses and their revenue trends.

4. Business Recommendations

- Summarize key insights from your analysis.
- Provide actionable recommendations for casino operators.
- Suggest whether casinos should increase or decrease promotional spending.
- Identify risk factors that could impact revenue stability.

Submission Guidelines

- Your final report should be **well-documented** and structured.
- You can submit your work as a **Jupyter Notebook (IPYNB) with explanations** and **detailed PPT report**.
- Ensure that all code, visualizations, and interpretations are included.